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Psychologist Steven Pinker argues that people have an intuitive sense of what's known to be known, even when they don't say it explicitly.

How do you know what I know you know? Steven Pinker on common knowledge

From using words to overthrowing regimes, shared awareness is necessary for people to coordinate. The cognitive scientist explains to *Nature* how it works.

How do we know when others know what we know? Harvard psychologist Steven Pinker's latest book, *When Everyone Knows That Everyone Knows*, delves into how 'common knowledge' can cement or explode social relations.

Common knowledge — awareness of mutual understanding — can explain the emergence of social-media shaming mobs, academic cancel culture and revolutions that seem to erupt from nowhere. It drives how people coordinate with others and can explain everything from awkward first-date conversations to financial bubbles and stock-market crashes.

Pinker tells *Nature* why it helps to better understand the ways we get into each other's heads — and what happens when we know that we have.

What is common knowledge?

It is the state in which I know something, you know it, I know that you know it, you know that I know it, I know that you know that I know that you know it, and so on, ad infinitum. It differs from private knowledge, in which someone knows something without knowing whether anyone knows they know it.

That's a lot of 'knowing'!

Common knowledge is usually tacit, and it can be generated at a stroke when we witness something that is conspicuous, public or self-evident — I see something while seeing you see it, and vice versa. From there, we can deduce that each observer knows that the other knows it, and so on, to as many levels as we care to reel out — a process that cognitive psychologists call recursive mentalizing.

Why does it matter?

Common knowledge is necessary for coordination: two or more people making arbitrary choices that benefit them, as long as they make the same choice. A classic example is a rendezvous. It's not enough to know that your friend likes to go to Starbucks and so to head there, because he might know that you like to go to Pret A Manger and head there. It's not even enough to know that he knows that you

know that he likes Starbucks, and head there, because he knows that you know that he knows that you like Pret, which is where he might go.

Nothing short of common knowledge will get you to the same place at the same time. The obvious means of achieving this is a phone call, because direct speech is an excellent common-knowledge generator. The next best thing is common salience. The friends might gravitate to a nearby landmark, such as Nelson's Column in London's Trafalgar Square, because each can anticipate that it would pop into the mind of the other.

What other examples are there?

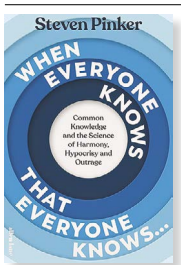
Our conventions depend on common knowledge, starting with language. The word 'rose' refers to a rose for no reason other than that everyone knows that everyone knows it does. Another convention is driving on the left rather than the right, or vice versa: there's no reason to choose one side or the other, but every reason to choose the same side as everyone else. This convention is backed up by law, but that doesn't matter: people have an incentive to follow the convention as long as they know that everyone else is following it.

Even legally binding conventions depend more on common knowledge than on enforcement, because no government has enough snitches and rooftop snipers to enforce every law. When 'no smoking' signs were first put up, people didn't report violators to the smoking police, but they could stand their ground when asking a smoker to stop and expect them to give way, because the smoker expected the complainant to stand their ground.

This is also how we can have social and moral norms, such as not telling ethnic jokes, revealing certain body parts in public or insulting someone's looks. The norms can persist by the common expectation that everyone knows that everyone knows there are certain things decent people just don't do.

How can common knowledge fail?

When there's a public signal that a belief might no longer be commonly held, the doubt can be self-fulfilling. In a bank run, a rumour that the bank might not have enough reserves to cover withdrawals can drive people to withdraw their savings out of fear that other people are withdrawing theirs out of the same fear. That can crash the bank, and, as the doubts reverberate, the economy.



When Everyone Knows That Everyone Knows...: Common Knowledge and the Science of Harmony, Hypocrisy and Outrage
Steven Pinker
Allen Lane (2025)

When then-US president Franklin D. Roosevelt said in 1933, "The only thing we have to fear is fear itself", he was expressing a theorem of common knowledge. Other runaway phenomena, such as financial bubbles or hyperinflation — or even the hoarding of toilet paper during a pandemic — are also driven by common expectation.

These are old ideas, so why write the book?

I was led into the study of common knowledge by my interest in language. A lot of speech can't be interpreted at face value. For example, "If you could pass the salt, that would be brilliant" has a veiled imperative meaning: "Give me the salt." I wanted to know why we don't all just say what we mean.

Common knowledge is what aligns us in social relationships. Our rituals of ignoring the elephant in the room, not blurting out everything we think, framing criticism tactfully, and other exercises of social skill consist of judgements of whether to make some fact common knowledge or whether to avoid doing so.

Do you include any fresh ideas?

I advance a hypothesis — which has driven my research over the past 15 years — that social relationships (such as friendship, romance, authority and transactional partnerships) are coordination games, ratified by common knowledge signals such as eye contact, laughter, dominance displays and direct speech.

And I address the cognitive question of what goes on in people's heads when they have common knowledge, given that multiple nested propositions about beliefs are cognitively strenuous.

And what does go on?

We intuitively grasp the existence of common knowledge as something that's public, or 'out there' or self-evident. And we talk about it by using metaphors for a conspicuous object, such as 'the elephant in the room'. This helps us to confidently coordinate with others who share that common knowledge when we deal with complex social situations.

Is there evidence for this idea?

My students and I tested the basic idea that a public signal emboldens people to coordinate by having people imagine themselves in a variant of Swiss philosopher Jean-Jacques Rousseau's classic game-theory dilemma of a stag hunt. Two incommunicado hunters must choose whether to take a chance on meeting up to chase large quarry (with the risk of going home empty-handed if the other party does not) or to safely pursue smaller quarry individually; only common knowledge makes it rational to choose coordination.

Our participants were more likely to gamble on meeting up when the presence of the large quarry was conveyed by a public smoke signal than when it was communicated by a sequence of one-way messages.

In other experiments, we showed that public conspicuity can enable recursive mentalizing by asking participants about a scenario in which an event (for example, a car crash) was either witnessed by two people directly or learnt about through hearsay. Only in the case of the public event were participants confident that each of the characters knew that the other knew, and knew that, and so on.

You say authoritarian governments don't like common knowledge.

That's why they don't allow freedom of speech or assembly: they don't want people to pool their private knowledge of the regime's oppressiveness into coordination-enabling common knowledge. In the film *Gandhi* (1982), the title character says, "One hundred thousand Englishmen simply cannot control 350 million Indians, if those Indians refuse to cooperate". But people can only refuse to cooperate if they have common knowledge that everyone else is refusing; otherwise they fear being picked off one by one. A public protest or widely circulated manifesto, through which everyone is aware of everyone else's discontent, can embolden people to collectively storm a palace or bring the machinery of the state to a halt.

How can this be linked to the suppression of academic freedom?

Biologists love to tell the (apocryphal) story about the wife of a bishop who, on learning about Charles Darwin's theory of evolution, exclaimed, "My dear, descended from the apes! Let us hope it is not true, but if it is, let us pray that it will not become generally known." She could have said commonly known. Moral norms are held up by common knowledge, and people fear that those norms might be threatened when some fact that seems to undermine them becomes commonly known. In her case, the norm was living a moral life and the compromising fact was denial of divine creation.

Today there are many facts that people fear would undermine values such as equity, dignity and social justice if they became common knowledge. Hence the efforts to censor, fire, shout down, disinvite, retract, and, in a revealing metaphor, deplatform — to remove someone from a location designed to generate common knowledge.

Whether or not a viable case can be made for keeping a small number of particularly dangerous questions out of common knowledge, the default must be intellectual freedom.

Interview by David Adam.
This interview has been edited for length and clarity.